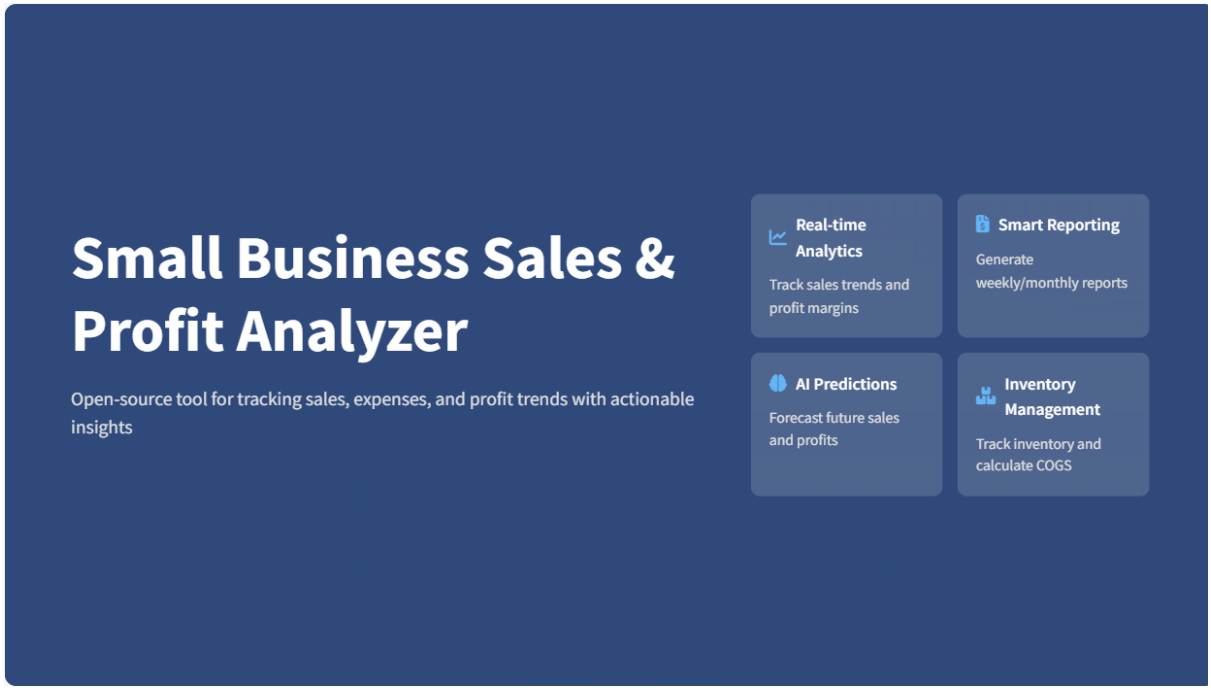
Small Business Sales & Profit Analyzer



Project Statement

Small and micro-business owners often struggle to track sales, expenses, and profit trends due to a lack of easy-to-use, affordable tools. While enterprise-level ERP systems exist, they are often too complex and costly for small shop owners, home-based businesses, or online sellers. This project aims to develop an **open-source, user-friendly tool** that allows business owners to log daily sales and expenses, automatically calculate profits, and generate weekly/monthly reports with clear visualizations. The system will include features for expense categorization, inventory adjustments, and predictive profit trends using the latest analytics techniques. By providing actionable insights in an accessible format, this tool will help small businesses make informed decisions, control costs, and maximize profitability.

Outcomes

- **Daily Sales & Expense Logging** – Simple interface for adding transactions quickly.
- **Automated Profit Calculation** – Calculates daily, weekly, and monthly profit.
- **Category-wise Analysis** – Breaks down income and expenses by category.
- **Inventory Adjustment Feature** – Optionally track inventory to link cost of goods sold (COGS).
- **Report Generation** – Auto-generated PDF/Excel reports for weekly and monthly summaries.

- **Predictive Trends** – AI-assisted prediction of future sales/profits based on historical data.
- **Data Visualization Dashboard** – Clear charts for sales trends, expense breakdown, and profit over time.
- **Secure User Accounts** – Multiple business profiles per user with role-based access control.

4. Modules to be Implemented

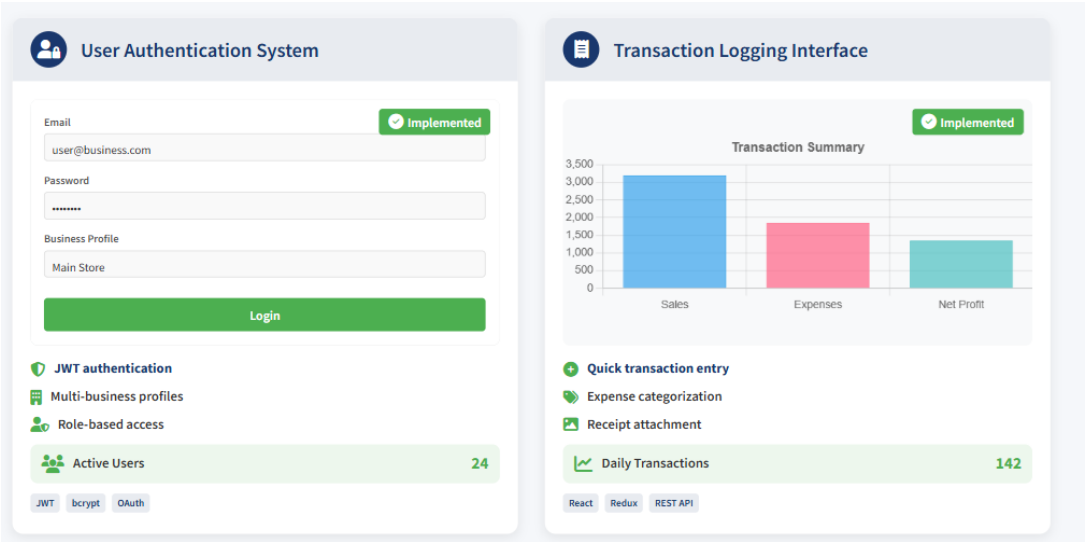
1. **User Authentication & Profile Management**
    - Secure registration and login (JWT authentication).
    - Profile creation for multiple businesses.
    - Role-based access (Owner, Accountant, Staff).
2. **Sales & Expense Management Module**
    - Add, edit, and delete transactions.
    - Categorize expenses (Rent, Utilities, Supplies, etc.).
    - Attach receipts or invoice images.
3. **Profit & Inventory Analysis Module**
    - Automatic profit calculation.
    - Optional inventory tracking for COGS calculation.
    - Alerts for low stock items.
4. **Analytics & Visualization Module**
    - Charts for sales trends, profit margins, and expense categories.
    - Weekly/monthly summary views.
    - AI-based forecasting for next month's sales and profit.
5. **Report Generation Module**
    - Export reports to PDF or Excel.
    - Email report scheduling.
6. **Admin Dashboard**
    - Manage business profiles and users.
    - Monitor usage and data quality.

- Configure system settings.

5. Week-wise Milestones

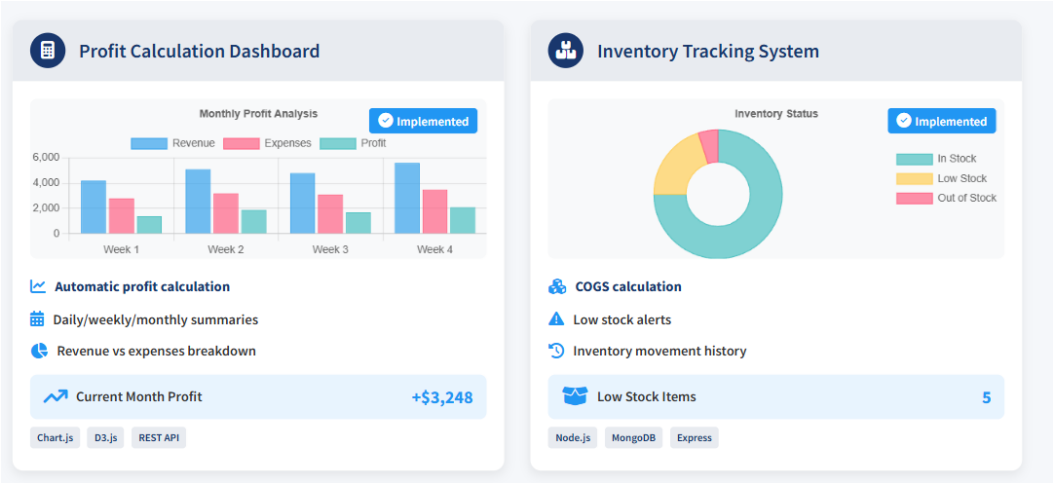
Milestone 1 (Weeks 1–2) – Authentication & Basic Transaction Logging

- Implement secure registration/login with JWT.
- Create profile management for businesses.
- Build basic sales/expense input forms.



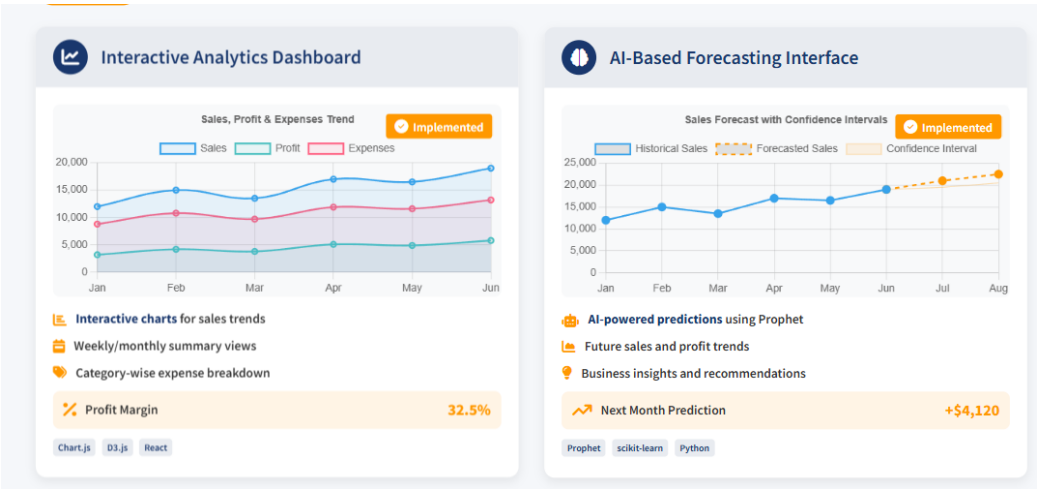
Milestone 2 (Weeks 3–4) – Core Profit & Inventory Tracking

- Implement profit calculation logic.
- Add inventory tracking with COGS computation.
- Simple dashboard with daily totals.



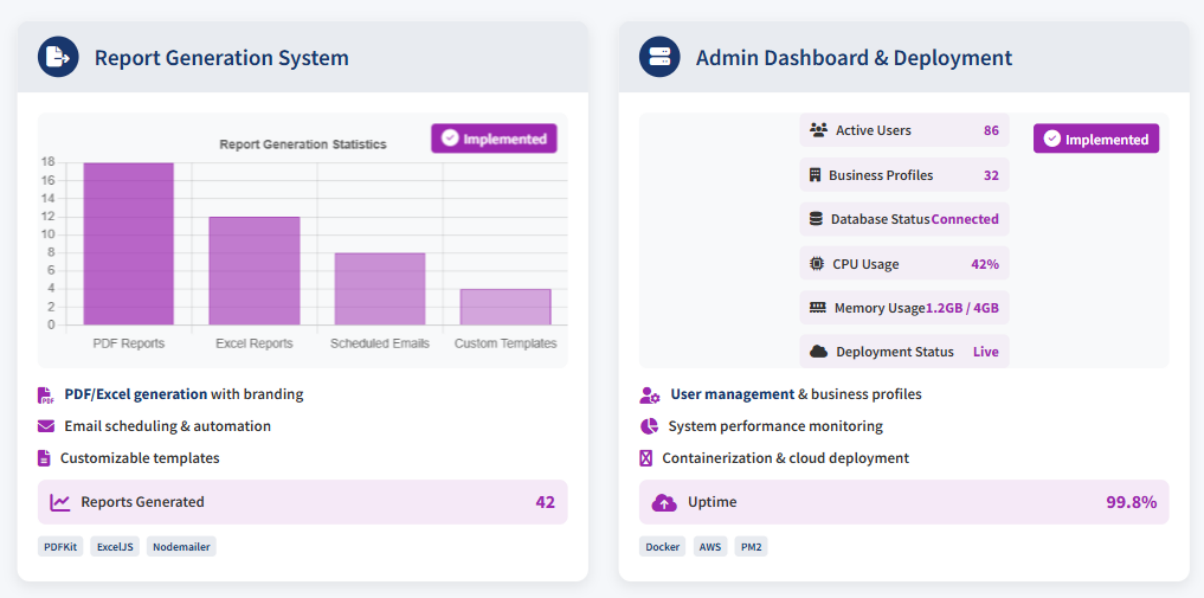
Milestone 3 (Weeks 5–6) – Advanced Analytics & Visualization

- Create interactive charts (sales, profit, expenses).
- Implement AI-based forecasting (e.g., Prophet or scikit-learn regression).
- Category-wise expense breakdown view.



Milestone 4 (Weeks 7–8) – Reports, Admin, and Deployment

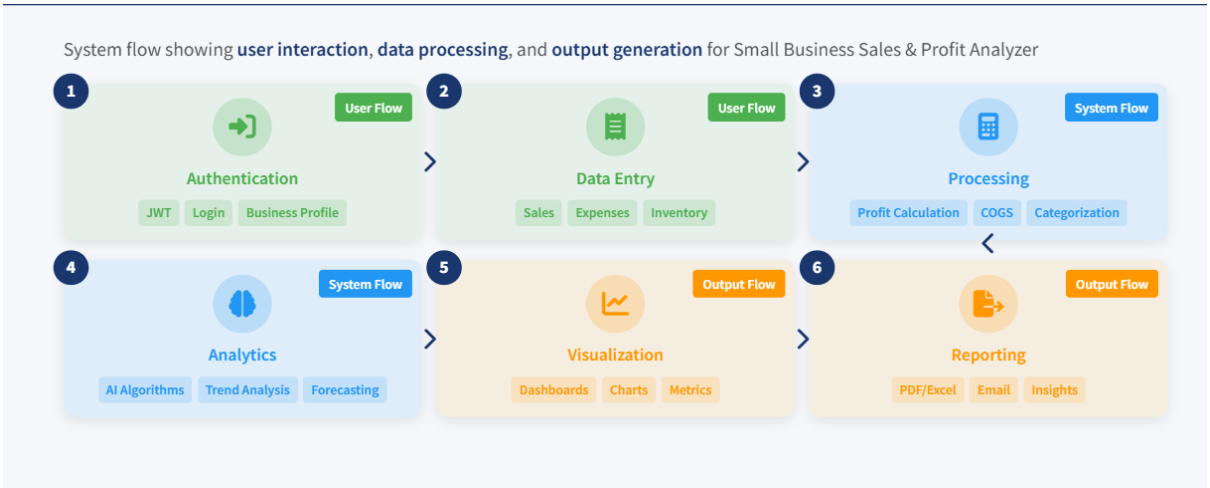
- Build PDF/Excel report generation.
- Create Admin dashboard for system monitoring.
- Containerize with Docker and deploy to free cloud hosting.
- Final documentation and presentation.



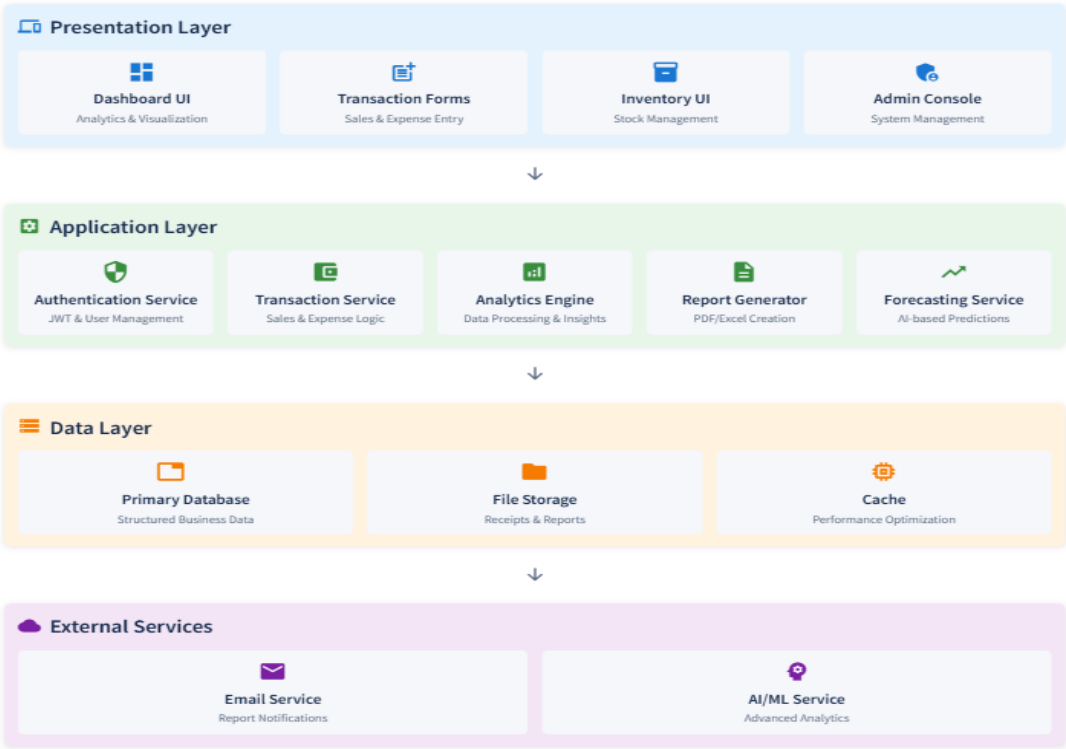
6. Evaluation Criteria

- **End of Week 2:** Functional authentication and basic transaction logging.
- **End of Week 4:** Working profit and inventory tracking.
- **End of Week 6:** Fully functional analytics dashboard with forecasting.
- **End of Week 8:** Report generation, admin control, and deployment complete.

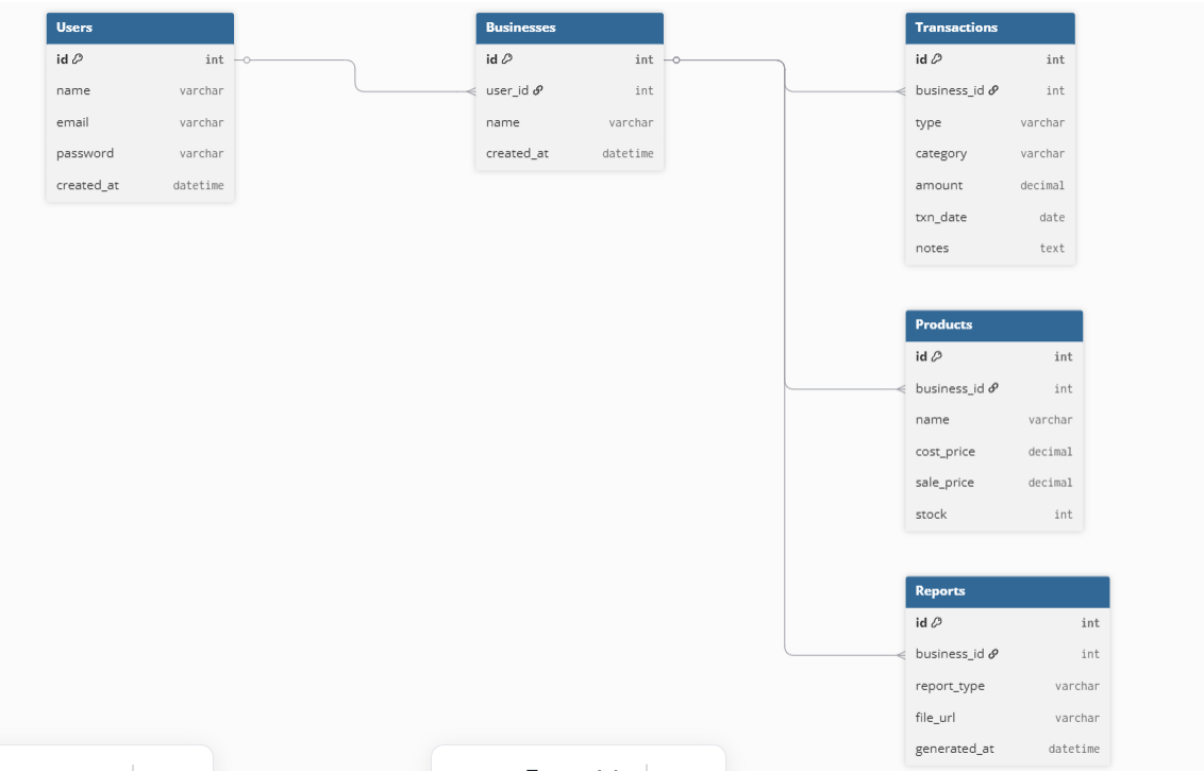
7. Work Flow Diagram



8. Architecture Diagram



9. Data Schema Diagram


Users
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name varchar
email varchar
password varchar
created_at datetime
Businesses
id int
user_id int
name varchar
created_at datetime
Transactions
id int
business_id int
type varchar
category varchar
amount decimal
txn_date date
notes text
Products
id int
business_id int
name varchar
cost_price decimal
sale_price decimal
stock int
Reports
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business_id int
report_type varchar
file_url varchar
generated_at datetime